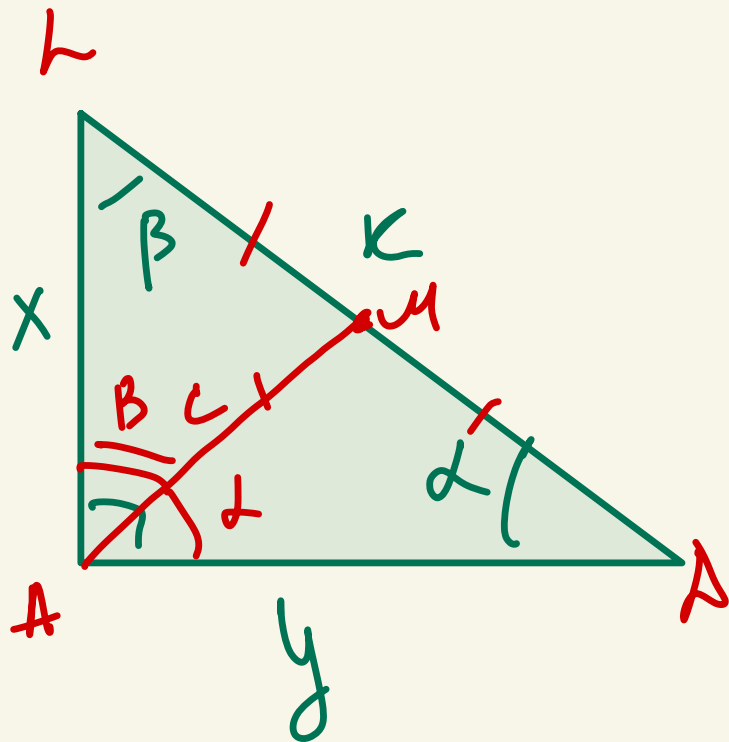






$$x = y \Rightarrow \alpha = \beta = 45^\circ$$

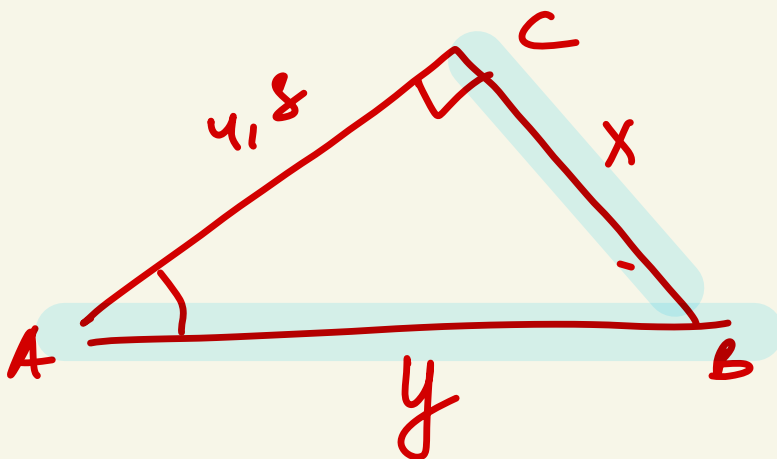
$$\begin{cases} \frac{x}{k} = \sin \alpha \leftarrow \\ \frac{y}{k} = \cos \alpha \leftarrow \\ \frac{x}{y} = \tan \alpha \end{cases}$$

$$x^2 + y^2 = k^2$$

АМ - медиана

$$\Rightarrow \frac{1}{2} \sin \alpha \sin \gamma$$

$$C = \frac{k}{2}$$



$$\sin A = \frac{4}{2.5}$$

AB?

$$\frac{CB}{AB} = \sin A = \frac{7}{25}$$

$$\left[\begin{array}{l} \frac{x}{y} = \frac{7}{25} \\ x^2 + 4,8^2 = y^2 \end{array} \right.$$

$$x = \frac{7}{25} y$$

$$\left(\frac{7}{25}y\right)^2 + 4,8^2 = y^2$$

$$\frac{49}{625}y^2 + \left(\frac{48}{10}\right)^2 = y^2$$

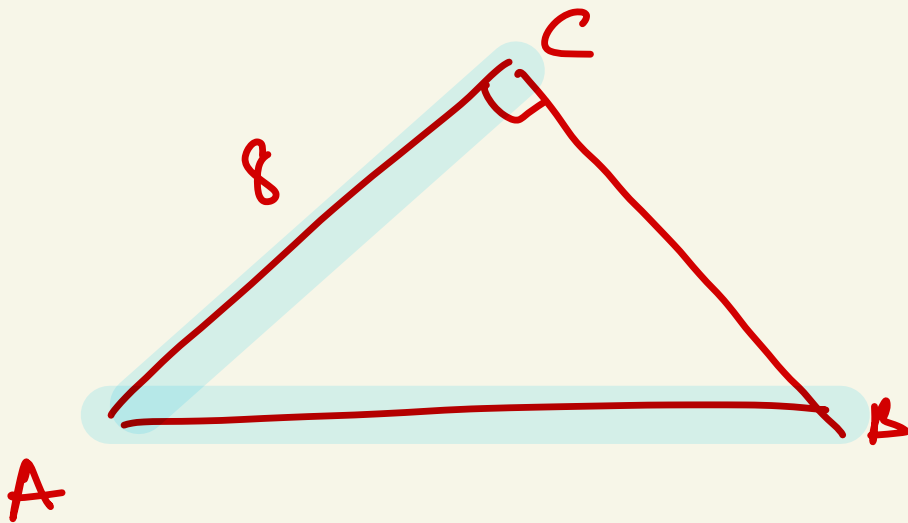
$$\frac{48^2}{100} = y^2 - \frac{49}{625}y^2$$

$$\frac{48^2}{100} = y^2 \left(\frac{625-49}{625} \right)$$

$$\frac{48^2}{100} = y^2 \frac{576}{625}$$

$$y^2 = \frac{625 \cdot 48^2}{100 \cdot 576} = \frac{25^2 \cdot 48^2}{10^2 \cdot 24^2}$$

$$y = \frac{25 \cdot 48}{10 \cdot 24} = 5$$



$$\cos A = \frac{2}{3}$$

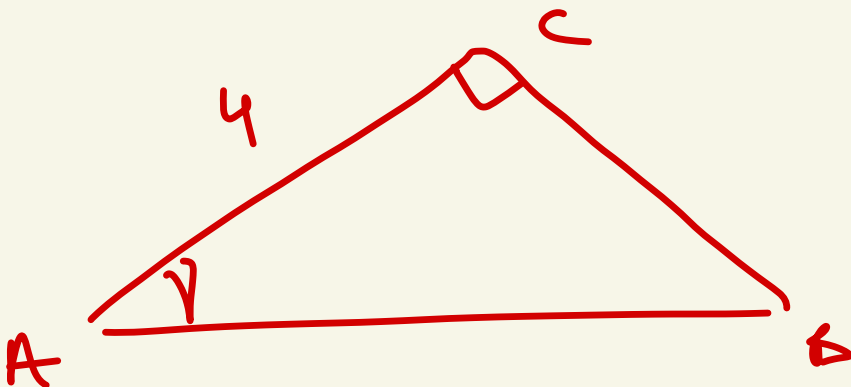
AB?

$$\frac{AC}{AB} = \frac{2}{3}$$

~~$$\frac{8}{AB} = \frac{2}{3}$$~~

$$\frac{3 \cdot 8}{2} = AB$$

$$AB = 12$$



$$\sin A = \frac{3}{5}$$

AB?

$$\left[\begin{aligned} \sin A &= \frac{CB}{BA} \\ 4^2 + CB^2 &= AB^2 \end{aligned} \right.$$

$$AB = 16 - \frac{3}{5} ?$$

$$\frac{CB}{AB} = \frac{3}{5}$$

$$CB = \frac{3}{5} AB$$

$$4^2 + \left(\frac{3}{5} AB\right)^2 = AB^2$$

$$16 + \frac{9}{25} AB^2 = AB^2$$

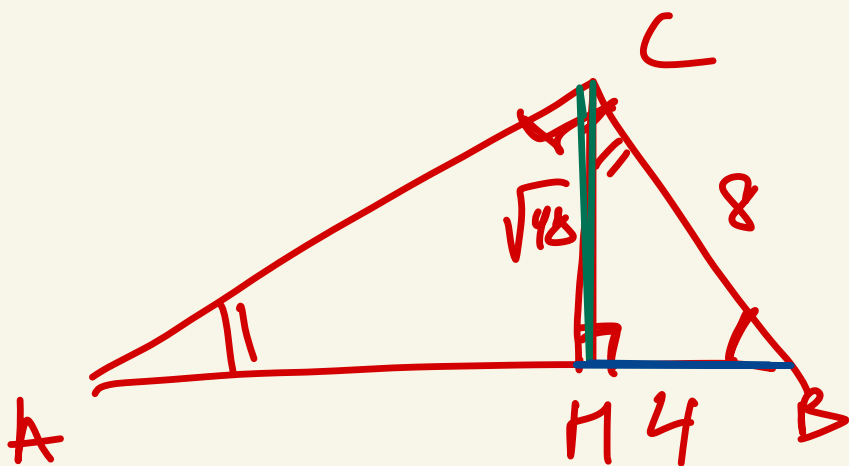
$$16 = AB^2 \left(1 - \frac{9}{25}\right)$$

$$16 = AB^2 \cdot \frac{16}{25}$$

$$\frac{16 \cdot 25}{16} = AB^2$$

$$AB^2 = 25$$

$$AB = 5$$



Sin A?

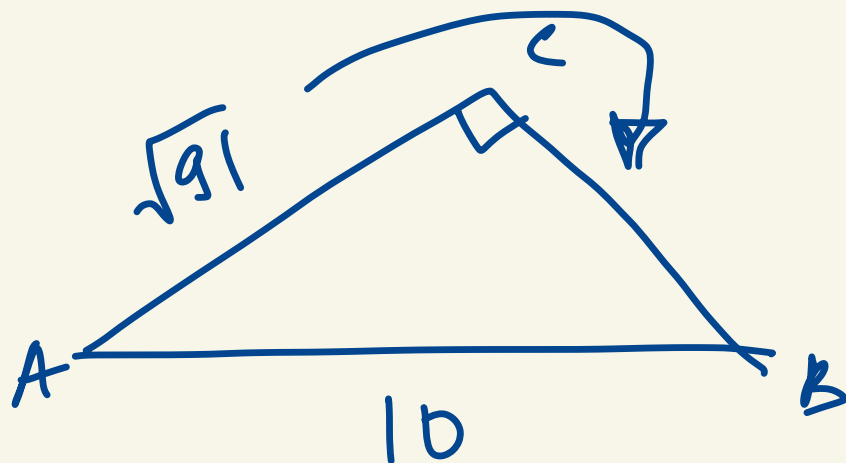
$$\sqrt{64 - 16} = \sqrt{48}$$

$$\frac{CH}{AB} ?$$

Δ ABC и Δ CHB
подобны

в подобных Δ
синусы, косинусы

$$\frac{1}{2} \cdot \frac{4}{8} = \frac{CH}{CB} = \sin C = \sin A$$

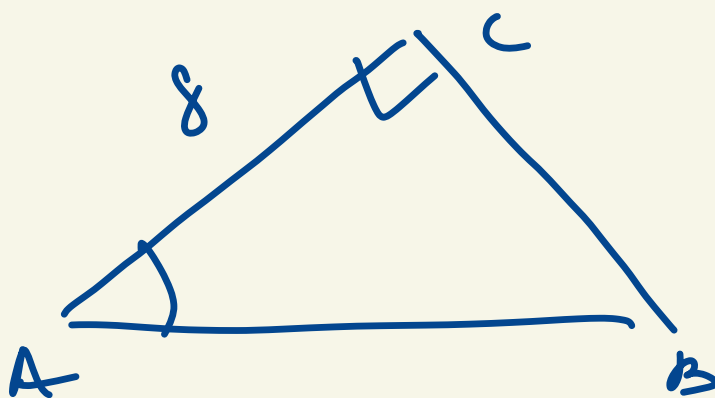


$$\begin{array}{|l} AB = 10 \\ AC = \sqrt{91} \end{array}$$

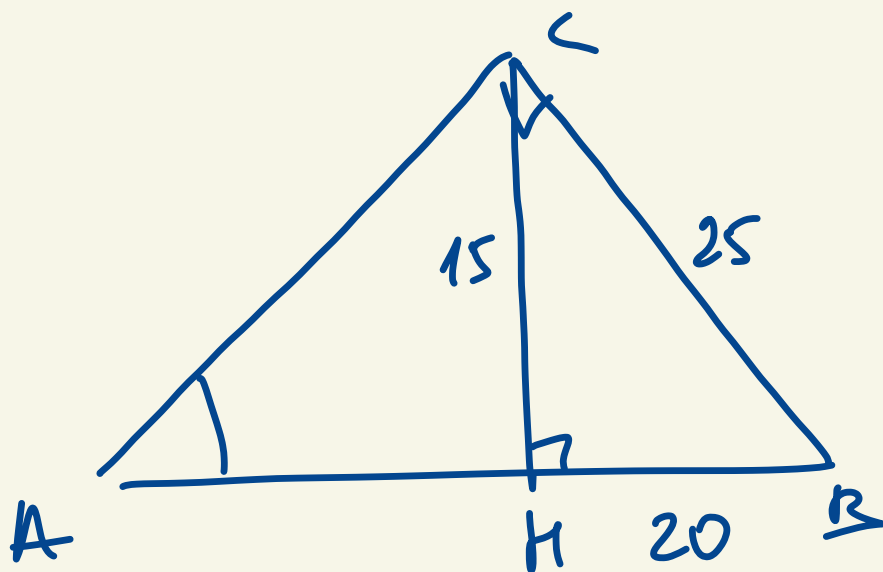
$$\sin A = ?$$

$$CB^2 = 100 - 91 = 9$$

$$CB = 3 \quad \rightarrow \quad \sin A = \frac{3}{10}$$



$$\begin{array}{l} AC = 8 \\ \tan A = 0,5 \\ BC \end{array}$$



$$\cos A = ?$$

$$\begin{array}{l} \angle A \rightarrow \angle C \\ \cos A = \cos C \end{array}$$

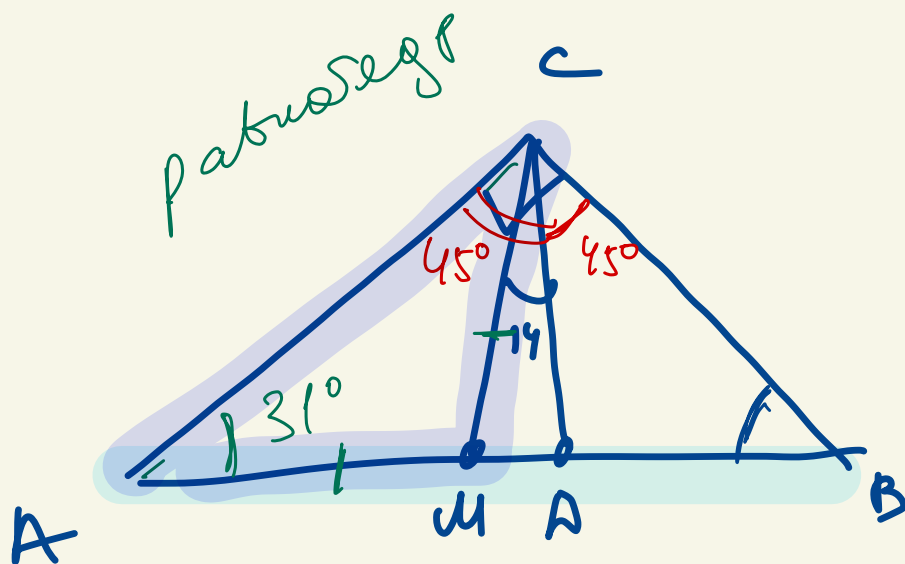
$$\cos \angle C = \frac{CH}{CB}$$

$$CH = 5?$$

$$CB^2 - HB^2 = CH^2$$

$$625 - 400 = 225$$

$$CH = 15$$



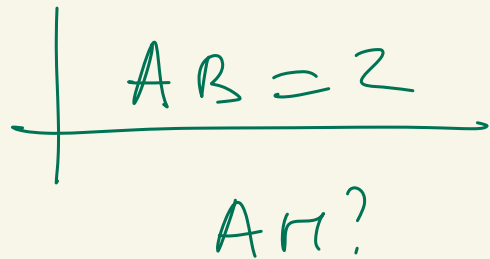
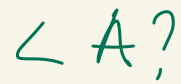
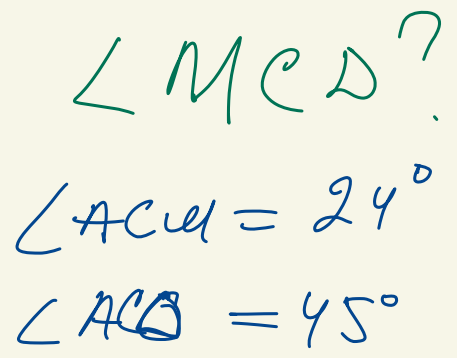
См - меди.
с А - бисс.

$\angle A?$

Σ

$$\angle ACH = 45 - 14 = 31^\circ$$

$$\angle ACB = 45^\circ$$



$$\frac{AH}{AC} = \cos 30^\circ$$

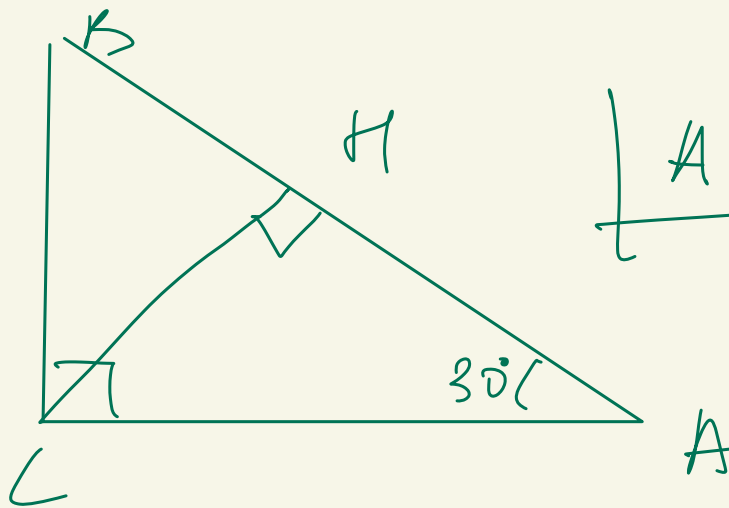
$$\frac{AC}{AB} = \cos 30^\circ$$

$$AC = AB \cos 30^\circ$$

$$AH = AC \cdot \cos 30^\circ = AB \cos 30^\circ \cdot \cos 30^\circ$$

$$\frac{\sqrt{3}}{2} \quad \frac{\sqrt{3}}{2}$$

$$2 \cdot \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} = \frac{3}{2} = 1,5$$



$$\frac{AB = 2\sqrt{3}}{CH?}$$

$$\frac{CH}{AC} = \sin 30^\circ$$

$$2\sqrt{3} \cdot \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = 2\sqrt{3} \cdot \frac{2 \cdot 3}{4} = \frac{6}{4} = \frac{3}{2}$$

BC: 10:00